

SIMATS ENGINEERING

Course Name: Data Structure

Course Code: CSA0302

LAB EXPERIMENTS 1-10

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1)

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int arr[100], n, i;
6     int *ptr;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter elements:\n");
12    for(i = 0; i < n; i++)
13        scanf("%d", &arr[i]);
14
15    ptr = arr;
16
17    printf("\nUsing Array Indexing:\n");
18    for(i = 0; i < n; i++)
19        printf("arr[%d] = %d\n", i, arr[i]);
20
21    printf("\nUsing Pointer Notation:\n");
```

INPUT

4
1 2 3 4

OUTPUT

Enter number of elements: Enter elements:

Using Array Indexing:

arr[0] = 1

arr[1] = 2

arr[2] = 3

arr[3] = 4

Using Pointer Notation:

*(ptr + 0) = 1

2)

QUESTIONS 8 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion.
Sample Input: Enter a number- 6
Sample Output: Factorial of 6 is 720
"

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int fact=1,i,n;
4     scanf("%d",&n);
5     for(i=1;i<=n;i++){
6         fact=fact*i;
7     }
8     printf("Factorial is %d",fact);
9 }
```

INPUT

6

OUTPUT

```
Factorial is 720
```

3)

QUESTIONS 8 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input (Matrix 1):
3 3 3

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 struct Triple
4 {
5     int row, col, value;
6 };
7
8 int main()
9 {
10     struct Triple A[100], B[100], C[200];
11     int r, c, n1, n2;
12     int i, j, k = 0;
13
14     printf("Enter rows, columns and non-zero elements: ");
15     scanf("%d%d%d", &r, &c, &n1);
16
17     printf("Enter row column value for Matrix 1:\n");
18     for (i = 0; i < n1; i++)
19         scanf("%d%d%d", &A[i].row, &A[i].col, &A[i].value);
20
21     printf("Enter rows, columns and non-zero elements: ");
```

INPUT

2 2 4
1 2
3 4

OUTPUT

```
Enter rows, columns and non-zero elements: Enter row column value for Matrix 1:
Enter rows, columns and non-zero elements: Enter row column value for Matrix 2:

Sparse Matrix after Addition:
Row Col Value
  2   3   4
0   0   0
0   0   0
```

4)

QUESTIONS 8 / 10 completed

Fibonacci Series

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series ($n \leq 20$) without using recursion. Start with 0 and 1.

Sample Input/Output:
Enter number of terms: 8
Fibonacci series: 0 1 1 2 3 5 8 13

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a=0,b=1,c,n,i;
4     scanf("%d",&n);
5     for(i=1;i<=n;i++){
6         printf("%d\\n",a);
7         c=a+b;
8         a=b;
9         b=c;
10    }
11 }
```

INPUT

15

OUTPUT

```
0
1
1
2
3
5
8
13
21
34
```

5)

QUESTIONS 8 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int i,a[100],n,pos;
4     scanf("%d",&n);
5     for(i=0;i<n;i++){
6         scanf("%d",&a[i]);
7     }
8     scanf("%d",&pos);
9     if(pos>=1 && pos<=n){
10         printf("the element is %d",a[pos-1]);
11     }
12     else{
13         printf("invalid position");
14     }
15 }
```

INPUT

6
1
2
3
4
3
7
5

OUTPUT

```
the element is 3
```

6)

QUESTIONS 8 / 10 completed

Fibonacci series (N ≤ 50)

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of the series and identify the largest term. Input validation is mandatory for N (1 ≤ N ≤ 50).

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int a=0,b=1,c,n,i;
4     scanf("%d",&n);
5     for(i=1;i<=n;i++){
6         printf("%d\n",a);
7         c=a+b;
8         a=b;
9         b=c;
10    }
11 }
```

INPUT

10

OUTPUT

```
0
1
1
2
3
5
8
13
21
34
```

7)

QUESTIONS 8 / 10 completed

Unsorted Array

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.

Sample Input: a=

[10,9,7,4,3,1,2]

Sample output: 5,6,8

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j;
6     int min, max, found;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter elements:\n");
12    for(i = 0; i < n; i++)
13        scanf("%d", &a[i]);
14
15    min = max = a[0];
16
17    for(i = 1; i < n; i++)
18    {
19        if(a[i] < min)
20            min = a[i];
21        if(a[i] > max)
```

INPUT

4 5 1 7 10

OUTPUT

```
Enter number of elements: Enter elements:
Missing elements: 2 3 4 6 8 9
```


8)

QUESTIONS 8 / 10 completed

Fib series-iterative ▾

FIB SERIES- ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include <stdio.h>
2 #include <time.h>
3
4 long long recursiveCalls = 0;
5 int fibRecursive(int n)
6 {
7     recursiveCalls++;
8
9     if (n == 0)
10         return 0;
11     if (n == 1)
12         return 1;
13
14     return fibRecursive(n - 1) + fibRecursive(n - 2);
15 }
16
17 int main()
18 {
19     int n, i;
20     int a = 0, b = 1, c;
21
```

INPUT

7

OUTPUT

OUTPUT

```
Enter N:
Iterative Fibonacci Series:
0 1 1 2 3 5 8

Recursive Fibonacci Series:
0 1 1 2 3 5 8

Execution Time (Iterative): 0.000 ms
Execution Time (Recursive): 0.000 ms
Total Recursive Calls: 59
```